Computational MR imaging
Laboratory 3: Partial Fourier imaging

Code submission is due by 23:55 on Monday. Please upload your code to StudOn in a described format. Late submissions will not be accepted.

Learning objectives

- Accelerate the acquisition of real-valued images using Hermitian symmetry in k-space
- Estimate the phase for partial Fourier reconstruction
- Apply the Margosian and POCS methods to reconstruct partial Fourier data

1. Zero-filled Fourier reconstruction and Hermitian symmetry reconstruction.

1.1. Kdata included in this exercise
- Fully sampled kdata: load_kdata_full()
- Partial Fourier sampled kdata: load_kdata_pf() with partial Fourier factor of 9/16

1.2. Define the half zero-filled kdata and the theoretical Hermitian symmetric kdata
- Implement method: *get_half_zf_kdata()*, which returns a half zero-filled kdata.
- Implement method: *hermitian_symmetry()*, which returns a theorical hermitian symmetric kdata of the half zero-filled kdata.
- Plot both kdata

1.3. Reconstruct the half zero-filled kdata and the Hermitian symmetry kdata.

1.4. Discuss artifacts for each case.

2. Phase estimation function

2.1. Implement method: *estim_phs()* to estimate the phase from a symmetric region at the center of k-space.

2.2. Estimate the phase of the zero-filled kdata.

2.3. Plot estimated phase

3. Margosian partial Fourier reconstruction

3.1. Implement method: *get_window()* which returns a window filter of either Ramp or Hamming.

3.2. Implement method: *pf_margosian()* for Margosian reconstruction.

3.3. Reconstruct PF kdata using Margosian method with both Ramp and Hamming window filters.

3.4. Discuss results between two filters in terms of SNR and potential artifacts.

4. POCS partial Fourier reconstruction

4.1. Implement method: *pf_pocs()* for POCS reconstruction.

4.2. Perform POCS reconstruction on the PF kdata with multiple iterations (2, 4, 6, 8, and 10)

4.3. Determine the minimum number of iterations at which the algorithm converges by comparing successive reconstructed images

5. Comparison of reconstructions

5.1. Plot the fully sampled reconstruction, zero-filled reconstruction, two Margosian reconstructions, and POCS reconstruction.

5.2. Compute differences between above reconstructions and the fully sampled reconstruction

5.3. Explain the reconstruction quality in terms of SNR, spatial resolution, residual artifacts and ringing artifacts.